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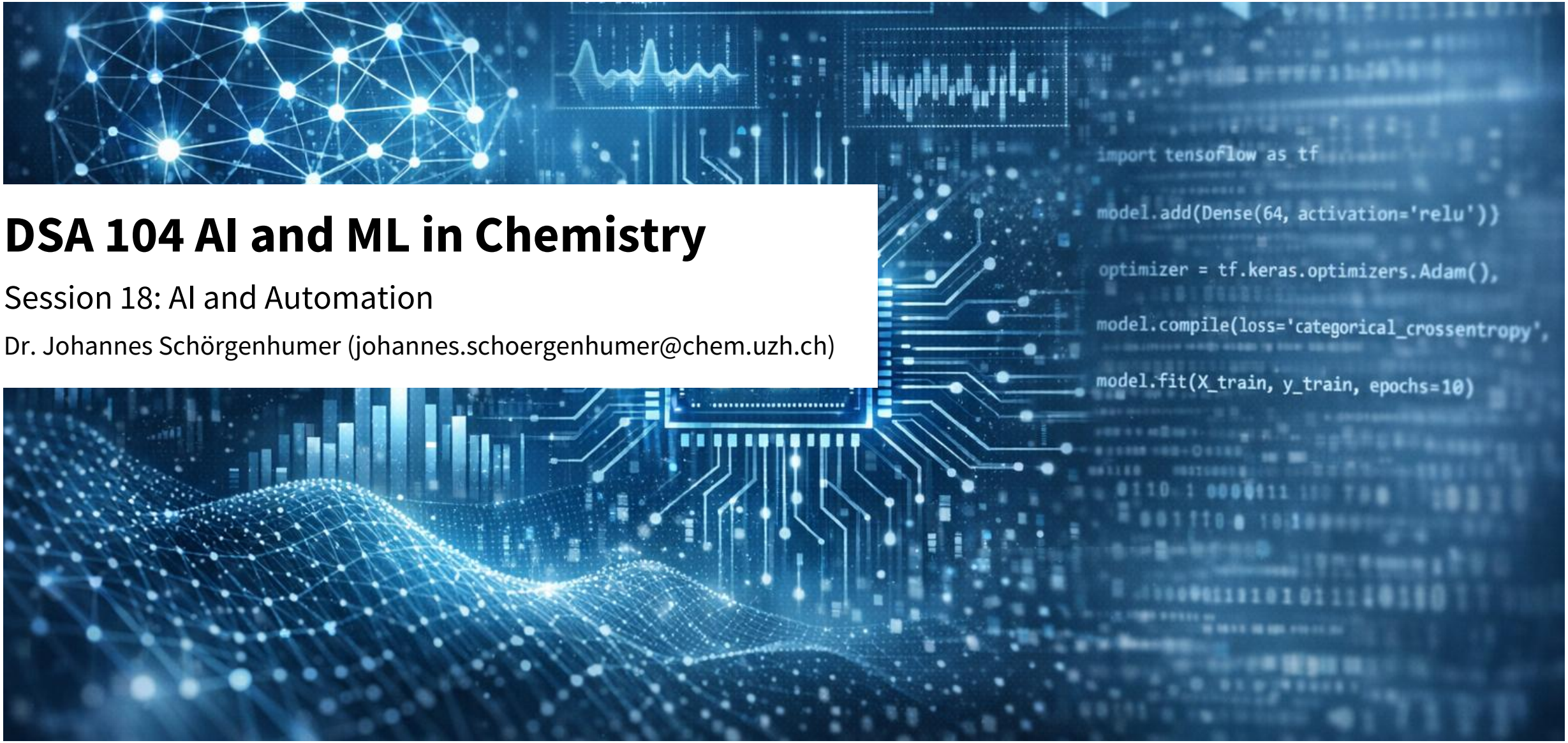
Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 18: AI and Automation

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

Points for today

Quick discussion assignment 7

Lab Automation in the context of AI

Exercise: Opentrons AI

Please register here:
https://ai.opentrons.com

14:00 Presentation by B12

Lab Automation supporting AI

“Data is the fuel of our time”

Good data is the basis of all ML/DL approaches!

— Reliable

— Reproducible

— Accessible

— Should be generally **FAIR**

Lab Automation is there to deliver this in the life science domain!

— **Highly reproducible** data (low random error)

— Higher throughput = **more data**

— **Standardised** processes

— Naturally intertwined with data science!

Talk by Joshua Sin (Roche) on May 27 (DSA Symposium, morning session)

https://www.go-fair.org/

Findable
Accessible
Interoperable
Reusable

Vice versa: AI supporting Lab Automation

Automated processes are complex and difficult to set up or control.

AI tools can provide assistance in:

- Experiment planning and design
- Plate design / screening plan
- Creating automated workflows
- Supervising / supporting the experiments:
    - Smart sensors / intelligent feedback systems
    - Error detection, e.g. by image recognition
    - Process control and calibration
- Controlling robotic tools

Automated Labs and AI: A powerful combination

Most modern Lab automation:

— **API** for data import/export = interface for ML

— **Integrated AI** tools (“**agents**”)

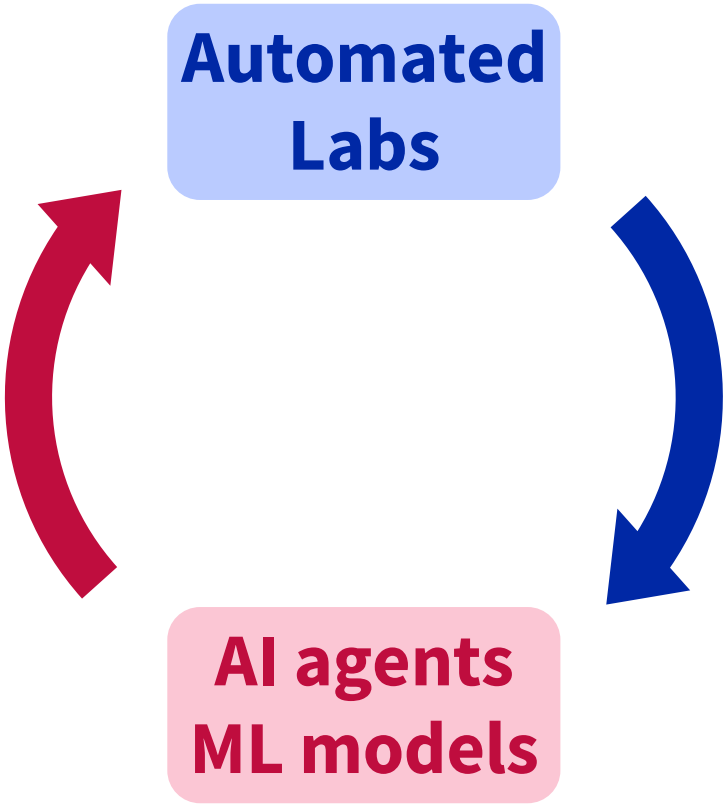
Closing the loop:

— minimising human intervention (deplacement)

— Hot topic: autonomous, self-driving research,…

Still mostly visionary and conceptual

Practical consideration: Is it the best solution?



b12

https://b12-labs.com/

ATINARY
ENABLING SELF-DRIVING LABORATORIES

https://atinary.com/

SciY

https://www.sciy.com/en.html

Autonomous Labs: Closed-loop processes

What are some pros and cons of SDLs?

DMTA cycle fully automated, decision processes and process control performed by AI tools

Trendy phrase: **S**elf-**D**riving **L**abs (**SDLs**)

Research tasks simplified:

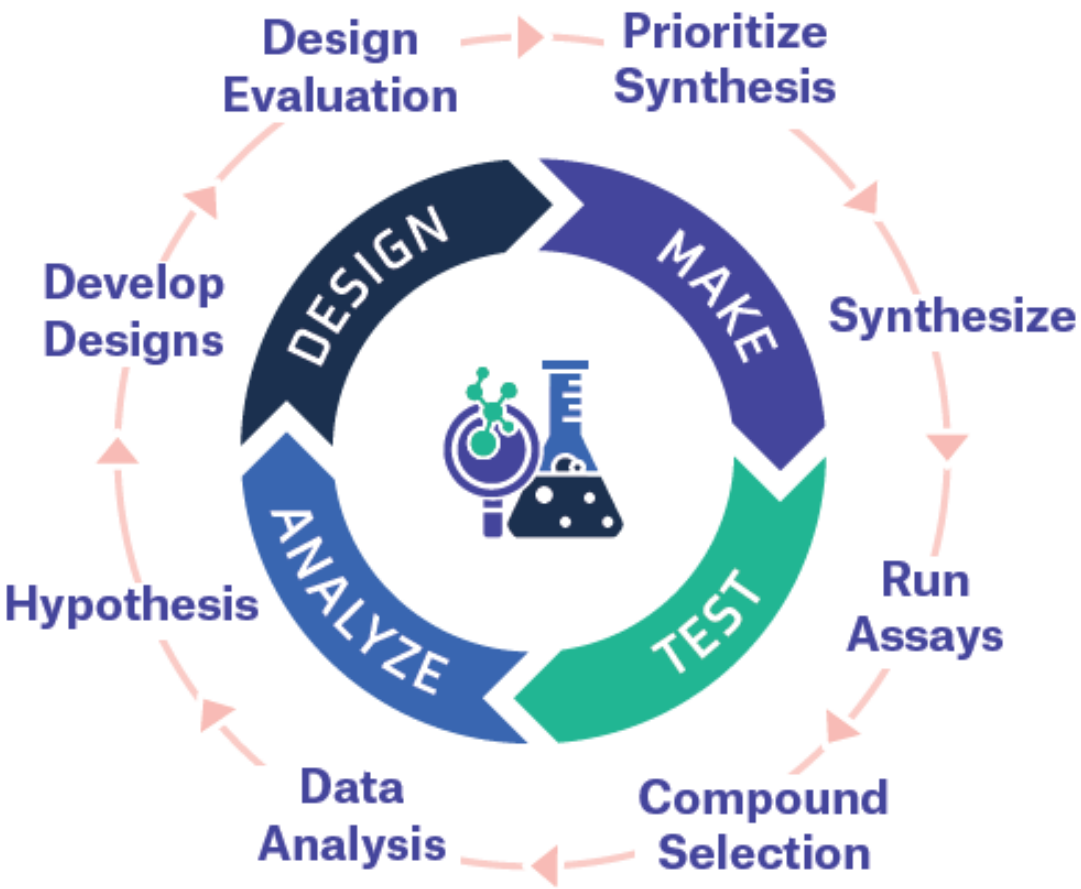
— from: “Let’s look at the target compound, review literature, test reference conditions, optimise for use case…”

— to: “Make the target compound and optimise the conditions.”

Examples:

— SwissCAT+ West at EPFL

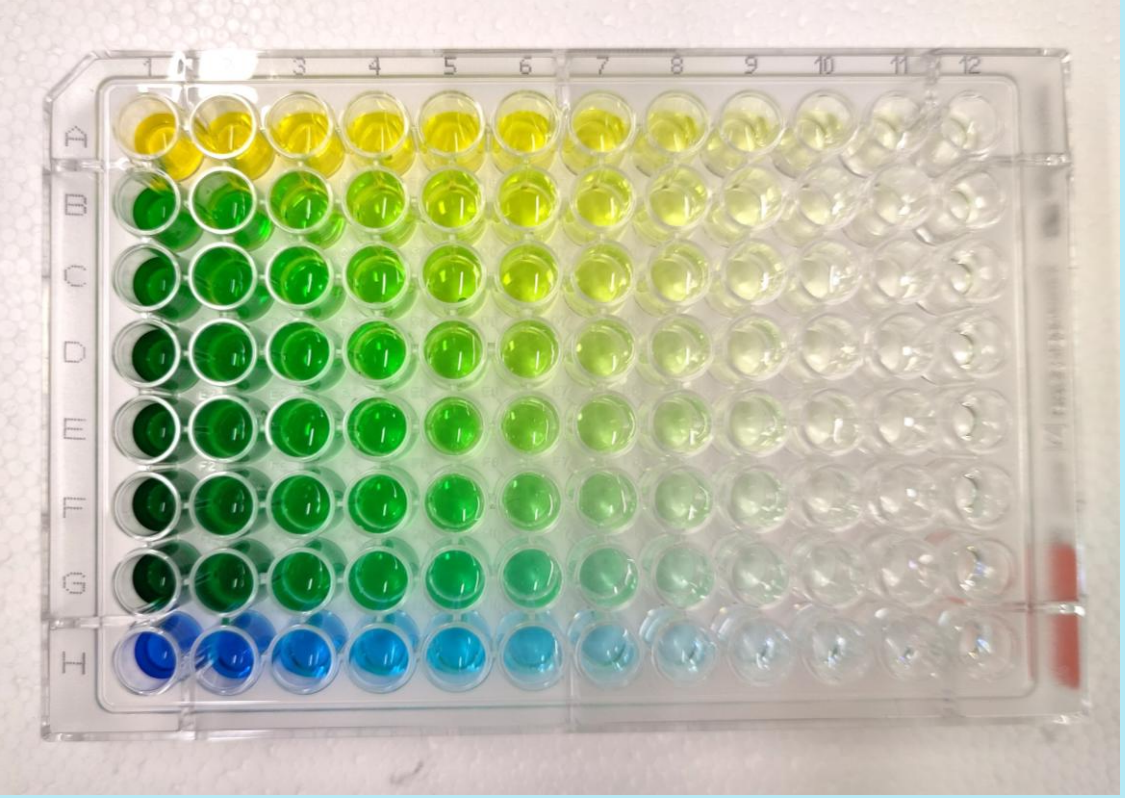
— SDLabs by Atinary (**Talk by Loïc Roch on May 27, DSA Symposium, afternoon session**)



Picture from: https://www.20visioneers15.com/post/strateos-digital-transformation-cloud-lab

Exercise: Create a protocol for the OT2 using the Opentrons AI

Produce a workflow for the following dilution series:



Boundary conditions:

— Gen2 pipettes: 1CH1000 and/or 8CH300

— Use opentrons labware from the official library

— Vertical: Two overlapping dilutions (two dyes in opposite direction – could be different ratios of two reagents)

— Horizontal: factorial dilution, last row = water as blank

— Make up volume: All wells need to have the same volume in the end.

Simulate the protocol in VS Code by running in the terminal:
opentrons_simulate.exe protocol_name.py

For this you need to activate the **specific environment** (in the opentrons-env folder in session 18) and navigate to the file (or add the path to it in the simulation command)

Self-driving labs Hackathon by EPFL LIAC and UZH HTEL

Interested in a new challenge?

Join the world's first practical lab automation hackathon!

How to participate:

1) Have a look at the announcement
2) Gather some teammates
3) Think of some cool research projects
4) Apply (same link as above) – deadline May 31
5) Just do it ☺
6) Win a prize?